

CERTIFICATE OF COMPLIANCE - RESIDENTIAL PERFORMANCE COMPLIANCE METHOD

Project Name: John Vo Duplex RES ADU  
Calculation Date/Time: 2024-10-16T23:32:40-07:00  
Calculation Description: John Vo Duplex Res-ADU  
Input File Name: 4357 Logan Ave.ribd22

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| 01   | 02     | 03        | 04          | 05      | 06         | 07          | 08    | 09         | 10       | 11              | 12   | 13          | 14               |
|------|--------|-----------|-------------|---------|------------|-------------|-------|------------|----------|-----------------|------|-------------|------------------|
| Name | Type   | Surface   | Orientation | Azimuth | Width (ft) | Height (ft) | Mult. | Area (ft²) | U-factor | U-factor Source | SHGC | SHGC Source | Exterior Shading |
| 25   | Window | Wall-Back | Back        | 90      | 3          | 4           | 1     | 12         | 0.3      | NFRC            | 0.23 | NFRC        | Bug Screen       |

| 01   | 02               | 03         | 04       |
|------|------------------|------------|----------|
| Name | Side of Building | Area (ft²) | U-factor |
| 101  | Wall-Front       | 20         | 0.2      |
| 102  | Wall-Right       | 20         | 0.2      |

| 01                  | 02          | 03         | 04             | 05                            | 06                            | 07                | 08     |
|---------------------|-------------|------------|----------------|-------------------------------|-------------------------------|-------------------|--------|
| Name                | Zone        | Area (ft²) | Perimeter (ft) | Edge Insul. R-value and Depth | Edge Insul. R-value and Depth | Carpeted Fraction | Heated |
| Slab On Grade Floor | Conditioned | 640        | 104            | none                          | 0                             | 80%               | No     |

| 01                | 02             | 03                | 04                 | 05                   | 06                                     | 07       | 08                                                                                                                                    |
|-------------------|----------------|-------------------|--------------------|----------------------|----------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------|
| Construction Name | Surface Type   | Construction Type | Framing            | Total Cavity R-value | Interior / Exterior Continuous R-value | U-factor | Assembly Layers                                                                                                                       |
| R15 Ext Wall      | Exterior Walls | Wood Framed Wall  | 2x4 @ 16 in. O. C. | R-15                 | None / 15                              | 0.036    | Inside Finish: Gypsum Board<br>Cavity / Frame: R-15 / 2x4<br>Sheathing / Insulation: R-15 Sheathing<br>Exterior Finish: 3 Coat Stucco |

Registration Number: 424-P010250939A-000-000-0000000-0000  
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CA Building Energy Efficiency Standards - 2022 Residential Compliance

Registration Date/Time: 10/16/2024 23:33  
Report Version: 2022.0.000  
Schema Version: rev 20220901

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| 01                | 02                     | 03                  | 04                                       | 05                   | 06                                     | 07       | 08                                                                                                                                                                 |
|-------------------|------------------------|---------------------|------------------------------------------|----------------------|----------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Construction Name | Surface Type           | Construction Type   | Framing                                  | Total Cavity R-value | Interior / Exterior Continuous R-value | U-factor | Assembly Layers                                                                                                                                                    |
| ROOF Asphalt      | Attic Roofs            | Wood Framed Ceiling | 2x4 @ 16 in. O. C.                       | R-0                  | None / None                            | 0.484    | Roofing: Light Roof (Asphalt Shingle)<br>Roof Deck: Wood<br>Siding/sheathing/decking<br>Cavity / Frame: no insul. / 2x4<br>Inside Finish: Gypsum Board             |
| Ceiling R30       | Ceilings (below attic) | Wood Framed Ceiling | 2x4 Bottom Chord of Truss @ 24 in. O. C. | R-0                  | 30 / None                              | 0.03     | Attic Floor: Wood<br>Siding/sheathing/decking<br>Cavity / Frame: no insul. / 2x4 Btm Chrd<br>Sheathing / Insulation: R-30 Sheathing<br>Inside Finish: Gypsum Board |

| 01                                    | 02                                 | 03                            | 04    | 05    |
|---------------------------------------|------------------------------------|-------------------------------|-------|-------|
| Quality Insulation Installation (QII) | High R-value Spray Foam Insulation | Building Envelope Air Leakage | CFM50 | CFM50 |
| Required                              | Not Required                       | N/A                           | n/a   | n/a   |

| 01         | 02                       | 03                | 04                | 05              | 06                   | 07                   | 08                | 09                    |
|------------|--------------------------|-------------------|-------------------|-----------------|----------------------|----------------------|-------------------|-----------------------|
| Name       | System Type              | Distribution Type | Water Heater Name | Number of Units | Solar Heating System | Compact Distribution | HERS Verification | Water Heater Name (#) |
| DHW System | Domestic Hot Water (DHW) | Standard          | Water Heater      | 1               | n/a                  | Basic                | n/a               | Water Heater (1)      |

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|--------------|------------|-----------------|----------------------|----------------------|---------------|-----------------------|------------------------|
| Name         | # of Units | Tank Vol. (gal) | NEEA Heat Pump Brand | NEEA Heat Pump Model | Tank Location | Duct Inlet Air Source | Duct Outlet Air Source |
| Water Heater | 1          | 50              | A. O. Smith          | FPTU 50 120 (50 gal) | Outside       | Outside               | Outside                |

| 01                 | 02                        | 03                                                            | 04                                                        | 05                                                   | 06                 | 07                |
|--------------------|---------------------------|---------------------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------|--------------------|-------------------|
| Dwelling Unit type | Water Heating System Name | Master Bath distance of furthest fixture to Water Heater (ft) | Kitchen distance of furthest fixture to Water Heater (ft) | Furthest Third furthest fixture to Water Heater (ft) | Compactness Factor | HERS Verification |
| Dwelling           | DHW System                | n/a                                                           | n/a                                                       | n/a                                                  | 0.7                | n/a               |

| 01               | 02              | 03              | 04                   | 05                        | 06                    | 07                               |
|------------------|-----------------|-----------------|----------------------|---------------------------|-----------------------|----------------------------------|
| Name             | Pipe Insulation | Parallel Piping | Compact Distribution | Compact Distribution Type | Recirculation Control | Shower Drain Water Heat Recovery |
| DHW System - 1/1 | Not Required    | Not Required    | Not Required         | Basic                     | Not Required          | Not Required                     |

| 01   | 02                               | 03                | 04                      | 05                | 06                      | 07          | 08                  | 09                       |
|------|----------------------------------|-------------------|-------------------------|-------------------|-------------------------|-------------|---------------------|--------------------------|
| Name | System Type                      | Heating Unit Name | Heating Equipment Count | Cooling Unit Name | Cooling Equipment Count | Fan Name    | Distribution Name   | Required Thermostat Type |
| HVAC | Heating and cooling system other | Heating System    | 1                       | Split System      | 1                       | Central Fan | Distribution System | Setback                  |

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| Name           | System Type         | Number of Units | Heating Efficiency | Heating Unit Brand |
| Heating System | Central gas furnace | 1               | AFUE - 80          | n/a                |

| 01           | 02               | 03              | 04                | 05                       | 06                    | 07                 | 08                    | 09                     |
|--------------|------------------|-----------------|-------------------|--------------------------|-----------------------|--------------------|-----------------------|------------------------|
| Name         | System Type      | Number of Units | Efficiency Metric | Efficiency EER/EER2/CEER | Efficiency SEER/SEER2 | Zonally Controlled | Mult-speed Compressor | HERS Verification      |
| Split System | Central split AC | 1               | EER/SEER          | 12                       | 14                    | Not Zonal          | Single Speed          | Split System-hers-cool |

| 01                     | 02               | 03             | 04                | 05                 | 06                          |
|------------------------|------------------|----------------|-------------------|--------------------|-----------------------------|
| Name                   | Verified Airflow | Airflow Target | Verified EER/EER2 | Verified SEERSEER2 | Verified Refrigerant Charge |
| Split System-hers-cool | Required         | 350            | Not Required      | Not Required       | Required                    |

| 01                  | 02                  | 03           | 04                       | 05                   | 06                  | 07     | 08     | 09     | 10             | 11                | 12                            |
|---------------------|---------------------|--------------|--------------------------|----------------------|---------------------|--------|--------|--------|----------------|-------------------|-------------------------------|
| Name                | Type                | Design Type  | Duct Ins. R-value Supply | Duct Location Return | Surface Area Supply | Return | Supply | Return | Bypass Duct    | Duct Leakage      | HERS Verification             |
| Distribution System | Unconditioned attic | Non-Verified | R-6                      | R-6                  | Attic               | Attic  | n/a    | n/a    | No Bypass Duct | Sealed and Tested | Distribution System-hers-dist |

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|-------------------------------|---------------------------|-------------------------|------------------------|----------------------|--------------|---------------------|-------------------------|-------------------------------------------------|
| Name                          | Duct Leakage Verification | Duct Leakage Target (%) | Verified Duct Location | Verified Duct Design | Buried Ducts | Deeply Buried Ducts | Low-leakage Air Handler | Low Leakage Ducts Entirely in Conditioned Space |
| Distribution System-hers-dist | Yes                       | 5.0                     | Not Required           | Not Required         | Not Required | Credit not taken    | Not Required            | No                                              |

| 01          | 02       | 03                    | 04                   |
|-------------|----------|-----------------------|----------------------|
| Name        | Type     | Fan Power (Watts/CFM) | Name                 |
| Central Fan | HVAC Fan | 0.45                  | Central Fan-hers-fan |

| 01                   | 02                     | 03                                |
|----------------------|------------------------|-----------------------------------|
| Name                 | Verified Fan Watt Draw | Required Fan Efficacy (Watts/CFM) |
| Central Fan-hers-fan | Required               | 0.45                              |

| 01              | 02            | 03                   | 04           | 05                             | 06                                    | 07                                | 08                | 09     |
|-----------------|---------------|----------------------|--------------|--------------------------------|---------------------------------------|-----------------------------------|-------------------|--------|
| Dwelling Unit   | Airflow (CFM) | Fan Efficacy (W/CFM) | IAQ Fan Type | Includes Heat/Energy Recovery? | IAQ Recovery Effectiveness - SRE/ASRE | Includes Fault Indicator Display? | HERS Verification | Status |
| Sfam IAQVentRpt | 64            | 0.35                 | Exhaust      | No                             | n/a / n/a                             | No                                | Yes               |        |

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| DOCUMENTATION AUTHOR'S DECLARATION STATEMENT                                                                                                                                                                                                                                                                                             |                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| I, I certify that this Certificate of Compliance documentation is accurate and complete.                                                                                                                                                                                                                                                 |                                                         |
| Documentation Author Name:<br>Francis Ng                                                                                                                                                                                                                                                                                                 | Documentation Author Signature:<br><i>Francis Ng</i>    |
| Company:<br>Delta Design                                                                                                                                                                                                                                                                                                                 | Signature Date:<br>10/16/2024                           |
| Address:<br>9855 Black Mountain Road                                                                                                                                                                                                                                                                                                     | CEA/ HERS Certification Identification (If applicable): |
| City/State/Zip:<br>San Diego, CA 92126                                                                                                                                                                                                                                                                                                   | Phone:<br>(858) 216-4334                                |
| RESPONSIBLE PERSON'S DECLARATION STATEMENT                                                                                                                                                                                                                                                                                               |                                                         |
| I certify the following under penalty of perjury, under the laws of the State of California:                                                                                                                                                                                                                                             |                                                         |
| 1. I am eligible under Division 3 of the Business and Professions Code to accept responsibility for the building design identified on this Certificate of Compliance.                                                                                                                                                                    |                                                         |
| 2. I certify that the energy features and performance specifications identified on this Certificate of Compliance conform to the requirements of Title 24, Part 1 and Part 6 of the California Code of Regulations.                                                                                                                      |                                                         |
| 3. The building design features or system design features identified on this Certificate of Compliance are consistent with the information provided on other applicable compliance documents, worksheets, calculations, plans and specifications submitted to the enforcement agency for approval with this building permit application. |                                                         |
| Responsible Designer Name:<br>Francis Ng                                                                                                                                                                                                                                                                                                 | Responsible Designer Signature:<br><i>Francis Ng</i>    |
| Company:<br>Delta Design                                                                                                                                                                                                                                                                                                                 | Date Signed:<br>10/16/2024                              |
| Address:<br>9855 Black Mountain Road                                                                                                                                                                                                                                                                                                     | License:                                                |
| City/State/Zip:<br>San Diego, CA 92126                                                                                                                                                                                                                                                                                                   | Phone:<br>(858) 216-4334                                |

Digitally signed by California Home Energy Efficiency Rating Services (CHEERS). This digital signature is provided in order to secure the content of this registered document, and in no way implies Registration Provider responsibility for the accuracy of the information.

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| Revision Dates |          |
|----------------|----------|
| 1              | 08.15.24 |
| 2              | 01.01.25 |
| 3              |          |
| 4              |          |
| SHEET TITLE    |          |
| TITLE-24       |          |
| Unit-B         |          |
| Drawing Date   |          |
| 08.27.26       |          |
| Check By       |          |
| Checker        |          |
| Drawn By       |          |
| Author         |          |
| Scale          |          |
| Job Number     |          |
| 022225         |          |
| Sheet Number   |          |